VULTR CLOUD COMPUTE + SERVERLESS INFERENCE

FORENSISAFE

Air-Gapped Hostile Document & Disinformation Detonation Vault

Safely detonating weaponized whistleblower leaks and isolating zero-day exploits with zero host exposure.

01 / Ephemeral Sandbox

256MB Vultr Linux containers with read-only rootfs and tmpfs RAM execution.

02 / Llama 3.1 AI Agent

Vultr Serverless Inference analyzes eBPF syscalls and purges malicious shellcode.

03 / Tamper-Proof Dossier

SHA-256 cryptographic chain of custody certificates admissible for legal/press use.

THE GLOBAL THREAT LANDSCAPE

Weaponized Leaks Are Compromising Journalists & Sources

X Zero-Day Infiltration in Disguise

Whistleblower drops (PDFs, spreadsheets, ZIPs) often contain embedded reverse shells, Pegasus-style spyware, or C2 beacons designed to silently take over the recipient's laptop in <200ms.

X Host Machine Vulnerability

Traditional antivirus software relies on signature databases that completely miss novel zero-days and obfuscated in-memory execution streams.

X Reconnaissance & Source Deanonymization

Opening investigative links directly leaks the journalist's public IP, device fingerprint, and geographical location, compromising source confidentiality.

X Lack of Legal Chain of Custody

Newsrooms lack forensic evidence tools to prove that a document was untampered before publication, exposing reporters to legal and ombudsman challenges.

CHALLENGE 1 COMPLIANCE & INFRASTRUCTURE

How Vultr Powers ForensiSafe (Blast Radius Zero)

Vultr Cloud Compute

VM Orchestration & Host

High-performance Ubuntu VM hosts the Next.js/Node orchestration engine. Controls ephemeral container lifecycles, assigns 256MB memory cgroups quotas, and enforces read-only file systems.

Vultr Serverless Inference

Llama 3.1 70B Reasoning

Direct integration with api.vultrinference.com/v1. Evaluates raw eBPF system call telemetry, classifies threat levels (Clean / Suspicious / Critical), and sanitizes legitimate whistleblower disclosures.

Network Sinkholing

Zero Outbound Egress

Untrusted code executes with --network none. All outbound C2 reverse shells and telemetry beacons are trapped by local mock sinkholes, killing malicious PIDs via seccomp filter.

SYSTEM WORKFLOW & CAPABILITIES

Three Operational Security Patterns

PATTERN A: 01 // Ephemeral File Detonator

* Drag-and-drop suspect scripts (.py, .sh), PDFs, and macros.
* Real-time eBPF syscall telemetry (TRAPPED, BLOCKED, ALLOWED).
* The Containment Moment: instant alert showing host OS intact and container destroyed.

PATTERN B: 02 // Browser Airlock

* Inspects suspect disinformation links and phishing portals.
* Quarantines WebGL fingerprinting, canvas trackers, and malicious redirects.
* Protects reporter identity with zero IP leakage or cookie trail.

PATTERN C: 03 // Chain-of-Custody Dossier

* Generates immutable cryptographic audit logs for legal discovery.
* SHA-256 digital fingerprinting of artifacts and detonation logs.
* One-click PDF export certified by ForensiSafe on Vultr Cloud Compute.

DEPLOYMENT & SOCIAL IMPACT

Empowering the Free Press with Production-Ready Security

Live Project Deliverables

LIVE WEB APPLICATION
https://forensi-safe.vercel.app/

GITHUB REPOSITORY
https://github.com/swatiicfai/ForensiSafe

PRIMARY CHALLENGE TRACK
Challenge 01: Blast Radius Zero

VULTR AI MODEL USED
Llama 3.1 70B Instruct (Serverless)

BACKEND ORCHESTRATION
Vultr Cloud Compute (Docker + eBPF)

Judging Criteria Alignment

1. Application of Technology

Deep, non-trivial integration of Vultr Cloud Compute and Vultr Serverless Inference (Llama 3.1 70B) with real kernel isolation.

2. Originality & Innovation

Goes far beyond standard chat wrappers to deliver real, air-gapped container sandboxing and verifiable Blast Radius Zero defense.

3. Business Value & Social Impact

Protects high-risk investigative journalists, whistleblowers, and human rights defenders from state-sponsored surveillance.

4. Presentation & Polish

High-fidelity cyber OSINT dashboard, live eBPF telemetry feeds, and exportable Chain of Custody certificates.